

Lab Report Guidelines

Lab 3: Pattern Discovery from Cleaned YouTube Text (Frequent Word Co-occurrence & Contextual Insights)

1. Cover Page

2. Objective

Briefly explain the purpose of this lab.

3. Tools and Libraries Used

Mention the Python version and all relevant packages.

4. Dataset Description

List and describe the datasets used in this lab:

- cleaned_comments.csv – Preprocessed YouTube comments from Lab 2.
- cleaned_captions.csv – Preprocessed YouTube captions from Lab 2.

Mention that each file contains a cleaned_tokens column representing tokenized words.
Note any issues such as empty token lists or non-English text detected during data loading.

5. Methodology / Procedure

5.1 Setup & Data Recall

- Load cleaned_comments.csv and cleaned_captions.csv into pandas DataFrames.
- Verify the cleaned_tokens column.
- Drop missing or empty token rows.
- Print top 20 unigrams using collections.Counter.
- Install and verify mlxtend.

5.2 Transaction Construction

- Treat each token list as one transaction (basket).
- Remove baskets with fewer than 3 tokens.
- Assign unique transaction IDs.
- Plot histogram of basket lengths.
- Compute statistics (average, min, max basket length).

5.3 Manual Pattern Discovery

- Count pairwise word co-occurrences across baskets.
- Keep pairs with count ≥ 3 .

- Plot top 20 co-occurring pairs (horizontal bar chart).
- Compare with the unigram frequency plot.
- Visualize a network graph using networkx.
- Export co-occurrence data to CSV.

5.4 Algorithmic Pattern Discovery

- Convert transactions to one-hot encoded DataFrame.
- Apply Apriori with **min_support = 0.3, 0.2, and 0.1**.
- Filter itemsets of length 2 and 3.
- Generate association rules (confidence ≥ 0.6 , lift ≥ 1.2).
- Plot support vs confidence scatter.
- Save frequent itemsets and association rules as CSV files.

5.5 Exploration Using Variations

- Experiment with min_support = 0.15 and 0.05.
- Try stemming instead of lemmatization.
- Exclude tokens under 4 characters and rerun Apriori.
- Apply the pipeline on captions and merged datasets.
- Compare discovered patterns across these variations.

5.6 Pattern Interpretation & Visualization

- Plot top 10 2-itemsets (by support).
- Plot top 10 3-itemsets (by confidence).
- Create word cloud and cluster graph.
- Compare results for captions vs comments.
- Write three insight statements with support/confidence metrics.

6. Code Implementation

Provide clear, well-commented code for:

- Loading and cleaning data
- Building transactions
- Co-occurrence counting
- Apriori analysis
- Visualization and exporting results

7. Results and Discussion

Include screenshots, plots, or sample tables showing:

- Top unigrams and frequent pairs
- Co-occurrence network graph
- Frequent itemsets and association rules

- Support vs confidence plot

Discuss:

- Differences between manual and algorithmic patterns.
- Impact of changing min_support.
- Insights on topics emerging from YouTube data.
- Challenges faced (e.g., sparse data, redundant patterns).

8. Conclusion

Summarize your key findings.

9. Reflection

1. What surprising pair or rule did you find?
2. What was challenging about discovering patterns?
3. What would you explore next (e.g., sentiment, topic modeling)?

Submission Requirements

Please upload the following files:

1. Lab Report (PDF format)
2. Python code file (.py or .ipynb)
3. Processed datasets (frequent_itemsets.csv, association_rules.csv, etc.)
4. All generated plots (PNG format)

Deadline: Submit before the assigned due date.